

Effects of Repeated High-Intensity Effort Training or Repeated Sprint Training on Repeated High-Intensity Effort Ability and In-Game Performance in Professional Rugby Union Players

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Abstract

Glaise, P, Rogowski, I, and Martin, C. Effects of repeated high-intensity effort training or repeated sprint training on repeated high-intensity effort ability and in-game performance in professional rugby union players. *J Strength Cond Res* 38(5): 932–940, 2024—This study investigated the effects of repeated high-intensity efforts (RHIE) training compared with repeated sprint exercise (RSE) training on RHIE ability (RHIEa) and in-game performance in professional rugby union players. Thirty-nine, male, professional, rugby union players were randomly assigned to 3 training groups (RHIE training, RSE training, and control). Repeated high-intensity effort ability and high-intensity effort characteristics (including sprints, acceleration, and contact efforts) during official games were measured before and after a 10-week specific (RHIE, RSE, or control) training period. The results of this study showed that concerning RHIEa, both the RHIE and RSE training significantly increased the players' average sprint velocity ($p < 0.001$, $d = -0.39$ and $p < 0.001$, $d = -0.53$ respectively), average sled push velocity (ASPV; $p < 0.001$, $d = -0.81$ and $p = 0.017$, $d = -0.48$ respectively), and RHIE score ($p < 0.001$, $d = -0.72$ and $p < 0.001$, $d = -0.60$ respectively). Repeated high-intensity effort training trended in a smaller increase in average sprint velocity than RSE training, a larger increase in ASPV, and a similar increase in RHIE score. Concerning in-game high-intensity efforts, both the RHIE and RSE training produced significant improvements in the number of sprints ($p = 0.047$, $d = -0.28$ and $p < 0.001$, $d = -0.47$ respectively), total distance ($p < 0.001$, $d = -0.50$ and $p = 0.002$, $d = -0.38$ respectively), the number of accelerations ($p < 0.001$, $d = -0.37$ and $p = 0.003$, $d = -0.32$ respectively), and contact rate ($p < 0.001$, $d = -0.97$ and $p = 0.020$, $d = -0.28$ respectively). Conversely, the magnitude of the increase in contact rate was almost twice as high in RHIE compared with RSE training. To conclude, the findings of this study were that both RSE and RHIE training are effective methods for developing RHIEa and in-game high-intensity efforts in professional rugby union. In practical applications, as the gains in certain abilities and game performance data differed depending on the training method chosen, we suggest that coaches choose the most appropriate method according to the profile of the players, their position, and the style of play they want to develop.

Key Words: rugby conditioning, high-intensity efforts, GPS, athletic performance, contact rate

Introduction

In recent decades, the ability to repeat sprints has been considered a major determinant of performance in team sports (3,22,31). Repeated sprint ability (RSa) is defined as the ability to reproduce maximum or near-maximum intensity in brief efforts (<10 seconds), interspersed with short periods of incomplete recovery (usually <90 seconds) (22). Among team sports, rugby union is an opposition sport characterized by repeated high-intensity running and contact efforts, interspersed by active or passive recovery periods of random duration and intensity (38). The forwards are mostly involved in contact efforts (such as ruck, tackle, and maul) and cover short distances, whereas the backs produce high-intensity running (such as acceleration and sprint) and cover

large distances (29). In rugby union, RSa is also a key performance factor (13), as players with a high RSa perform more actions per time unit (37). In forwards, higher RSa levels are associated with more combat actions and accelerations (i.e., <3 m·s⁻²) in game, whereas in backs, RSa correlated with high-speed running (i.e., >21 km·h⁻¹) and sprint efforts (i.e., >25 km·h⁻¹) in game (23). Consequently, RSa is a determinant of in-game performance for all players in a rugby union team.

Rugby playing also requires contact efforts, which led to an evolution of the RSa concept named repeated high-intensity effort ability (RHIEa). The latter first appeared in rugby league (2), then in rugby sevens (8), and more recently in rugby union (36). Austin et al. (2) and Black et al. (4) defined repeated high-intensity effort (RHIE) as a sequence of 3 or more intense efforts interspersed with less than 21 seconds of recovery. In team sports, such as rugby union, RHIEa is considered a crucial element (25) and appears to be more relevant than RSa for assessing players' performance (26). Intense efforts include accelerations (≥ 2.79 m·s⁻²), high-speed running (> 5.0 m·s⁻¹), and contact efforts (e.g., tackles, rucks, or mauls) (1,2,20). In a game of rugby union, the

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occurrence and nature of RHIE bouts depend on the position of players. Forwards perform more RHIE (i.e., 11–21 bouts) compared with backs (2–18 bouts) in combat-based RHIE bouts, whereas backs produce more run-based RHIE bouts (2). More specifically, Sheehan et al. (36) demonstrated that the back rows perform more RHIE than other forwards (9 ± 5 RHIE per game), whereas centers and the back 3 (both 18 ± 6 RHIE per game) achieve more RHIE than other backs and the whole team. The average duration of an RHIE bout is between 26 ± 4 seconds (in inside backs) and 52 ± 7 seconds (in back rows) (2). Furthermore, in rugby league, Gabbett (17) showed that winning teams perform more RHIE bouts in the entire game, especially before conceding tries compared with losing teams that achieved more RHIE bouts before scoring tries. This suggests that high RHIE activity in winning teams also contributes to more solid defenses, which are harder to beat (19). These findings emphasize that RHIE bouts often occur at critical periods during games and confirm the importance of developing RHIEa in rugby union players.

Because of their different nature, and because contact efforts generate additional fatigue that strongly impacts running intensities (28) and tackle skill efficiency (18), RSa and RHIEa must be considered 2 distinct physical qualities; thus, they require specific training methodologies (26). Concerning repeated sprint exercise (RSE) training, linear sprint repetition is more effective than high-intensity aerobic interval training for aerobic and football-specific adaptations (14). Repeated shuttle sprints also efficiently improved RSa with a $2.61 \pm 2.8\%$ time decrease in a repeated shuttle-sprint ability test (7). In rugby union, the number of sprints performed during an RSa test was reported to increase by 64% after 7 RSE training sessions in hypoventilation-induced hypoxia (16). Although methods for improving RSa through RSE training have been extensively studied in team sports, including rugby union (3,6,7,16), additional knowledge is required concerning training methods aimed at improving RHIEa. Regarding RHIE training, Vachon et al. (39) recently demonstrated that RHIEa is improved by aerobic training coupled with speed training during high-intensity interval training

(HIIT) block training. These authors also show that pre-conditioning sequence, including contact, leads to a higher accumulation of fatigue compared with running preconditioning sequence, which impact the running intensities that could influence the development of RHIEa (39).

Despite these recent data, the effects of RHIE training on RHIEa and game performance remain insufficiently explored. In addition, high-intensity repetition training programs based on sprinting (RSE training) and on alternating high-intensity running and contact efforts (RHIE training) generate specific adaptations in RSa and RHIEa, respectively. However, the efficiency of these methods for developing RHIEa has never been compared, and whether these training methods differentially impact in-game efforts in a rugby union competitive situation remains unknown.

Therefore, this study aimed to determine whether RSE training and RHIE training differently modulate (a) RHIEa and (b) the production of high-intensity running and contact efforts in competitive game situations. We hypothesized that RSE training and RHIE training would be equivalently effective at improving RHIEa, with larger improvements for repeated sprints with RSE training, and also that RHIE training would more strongly improve repeated contact effort performance. Furthermore, we hypothesized that both training programs would produce similar improvements in game situations in terms of high-intensity running and contact efforts production.

Methods

Experimental Approach to the Problem

A 26-week interventional study was designed to compare the effects of RSE training and RHIE training focusing on RHIE ability and the production of high-intensity efforts in game. A test–retest procedure was applied over 26 weeks divided into 3 periods (Figure 1). The first period was the pretraining period from the first to the seventh weeks, during which the in-game

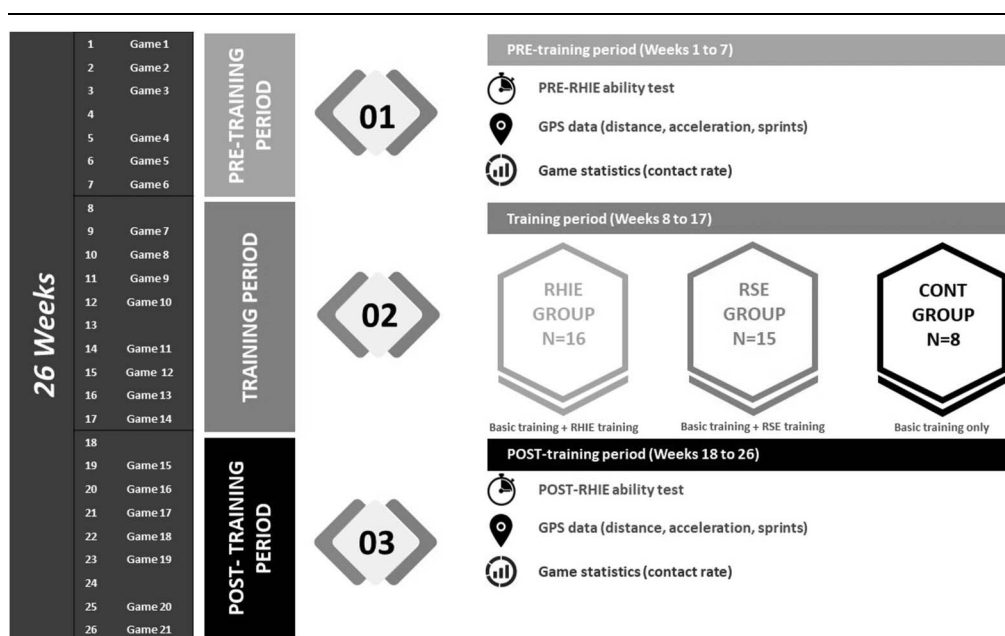


Figure 1. Study design.

performance was recorded based on 6 official competitive games, and the RHIEa was assessed between weeks 4 and 5. The second period lasted from the eighth to the 17th weeks and corresponded to the training period during which 10 sessions of specific training were performed (i.e., RSE, RHIE, or regular training). Finally, the third period lasted from the 18th to the 26th weeks and was the posttraining period, during which the in-game performance was recorded based on 7 official competitive games, and the RHIEa was assessed between weeks 17 and 19.

Subjects

Forty-five adult professional rugby union players playing in the second French National division (Pro D2 division) were initially included in the protocol. Five players were excluded from the study because of match injury and 1 for international selection. Thirty-nine subjects (age, 26.4 ± 4.2 years [19–33 years old]; height, 184.4 ± 7.6 cm; body mass, 97.2 ± 15.5 kg; four-skinfold body fat, $15.7 \pm 4.1\%$) completed the study and were randomly divided into the following 3 groups (Table 1): (a) the RHIE group ($n = 16$, with 9 forwards and 7 backs); (b) the RSE group ($n = 15$, with 8 forwards and 7 backs); and (c) the control (CONT) group ($n = 8$, with 5 forwards and 3 backs). Each participant achieved a training volume of approximately $20 \text{ h}\cdot\text{wk}^{-1}$ in 14 weekly sessions (8 rugby sessions, 4 strength and speed development sessions, 1 fitness development session, and 1 official game). Each participant agreed to participate in the study and signed an informed consent form. All the procedure was a part of the habitual team training program. The study received University of Lyon 1 institutional ethics approval and used safe procedures for the participants.

Procedures

Training Interventions. During the training period, additional specific sessions were implemented according to the training group as follows. Each participant from each group completed at least 7 of the 10 training sessions, and none of the subjects were unavailable for 3 consecutive weeks.

Players participated in the training intervention in addition to their regular training, which was composed of 8–14 training sessions per week divided into 3–6 conditioning sessions and 5–8 rugby team or line sessions per week. All descriptions of training interventions are detailed in Table 1.

To ensure that the training load was similar between the 3 groups, external load was measured during the training period with global positioning system (GPS) units (Catapult Innovations, Melbourne, Australia). The total achieved distance and the total PlayerLoad produced during the training period were monitored. Playerload is a variable calculated from GPS data, which includes accelerations, movements, and micromovements ($<2 \text{ m}\cdot\text{s}^{-1}$) and represents a quantity of work (external training load) (5).

$$\text{PlayerLoad} = \sqrt{\frac{(a_{y(t)} - a_{y(t-1)})^2 + (a_{x(t)} - a_{x(t-1)})^2 + (a_{z(t)} - a_{z(t-1)})^2}{100}}$$

The average playing time was also calculated as the total playing time per player during the whole study.

Repeated High-Intensity Effort Ability Assessment. The ability to repeat high-intensity efforts was evaluated through the RHIEa test (Figure 2). This test was adapted from the one

Table 1
Detailed intervention for RHIE and RSE training.*

	RHIE training	RSE training	CONT
Description	Players performed 10 training sessions (1 per week), which consisted of repeating 2 series of RHIE exercise in addition to the usual team training. Each RHIE series was composed of sets of 4–6 efforts of 3–6 s of high-intensity running (accelerations $>3 \text{ m}\cdot\text{s}^{-2}$ and high-velocity running $>5 \text{ m}\cdot\text{s}^{-1}$) and contact efforts (tackles, mauling tasks, and wrestling) with a start every 15–20 s (26,41)	Players performed 10 training sessions (1 per week), which consisted of repeating 2 series of RSE in addition to the usual team training. Each RSE series was composed of sets of 4–6 20-m maximal sprints with a start every 15–20 s (7,26)	Players did not receive any RHIE-specific or RSE-specific training and only completed the usual team training. During the RHIE and RSE training series, the CONT group participated in compensatory skill training drills
Training session			
1	2 × 4 high-intensity efforts started every 20-s cycle with passive recovery	2 × 4 sprints of 20 m started every 20-s cycle with passive recovery	
2	2 × 6 high-intensity efforts started every 20-s cycle with passive recovery	2 × 6 sprints of 20 m started every 20-s cycle with passive recovery	
3	2 × 4 high-intensity efforts start every 20-s cycle with active recovery	2 × 4 sprints of 20 m started every 20-s cycle with active recovery	
4	2 × 6 high-intensity efforts start every 20-s cycle with active recovery	2 × 6 sprints of 20 m started every 20-s cycle with active recovery	
5	2 × 6 high-intensity efforts start every 20-s cycle with active recovery	2 × 6 sprints of 20 m started every 20-s cycle with active recovery	
6	2 × 4 high-intensity efforts start every 15-s cycle with passive recovery	2 × 4 sprints of 20 m started every 15-s cycle with passive recovery	
7	2 × 6 high-intensity efforts start every 15-s cycle with passive recovery	2 × 6 sprints of 20 m started every 15-s cycle with passive recovery	
8	2 × 4 high-intensity efforts start every 15-s cycle with active recovery	2 × 4 sprints of 20 m started every 15-s cycle with active recovery	
9	2 × 6 high-intensity efforts start every 15-s cycle with active recovery	2 × 6 sprints of 20 m started every 15-s cycle with active recovery	
10	2 × 6 high-intensity efforts start every 15-s cycle with active recovery	2 × 6 sprints of 20 m started every 15-s cycle with active recovery	

*RHIE = repeated high-intensity effort; RSE = repeated sprint exercise; CONT = control.

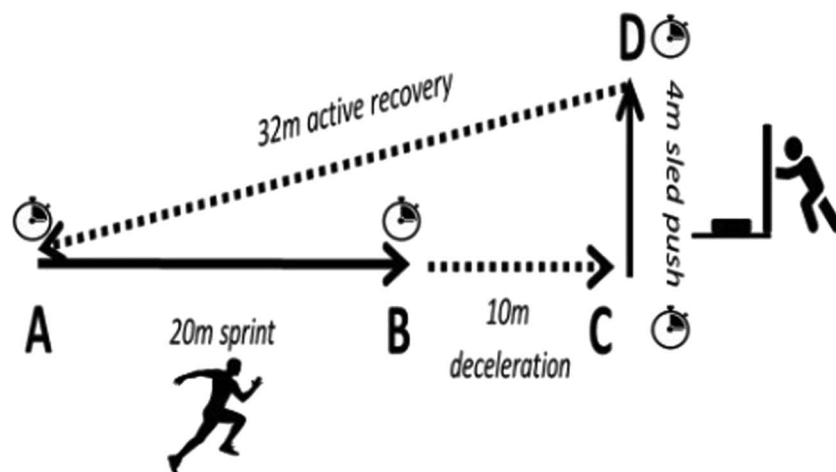


Figure 2. RHIE ability test. RHIE = repeated high-intensity effort.

proposed by Vachon et al. (41), which was derived from the test performed by Johnston and Gabbett (26). It consists of performing 10 repetitions of the following 30-second sequence: (a) a 20-m sprint at maximal velocity from a tripod starting position; (b) a 10-m deceleration (before the test, participants were instructed to perform the deceleration as strong as possible); (c) when the timer reaches 10 seconds, a 4-m sled push weighted at 60% of the player's body mass; and (d) a 32-m active recovery (low-intensity running $\geq 2 \text{ m}\cdot\text{s}^{-1}$) moving back to the start for the next sequence.

The test was performed on artificial turf in dry windless conditions. The times for the 20-m sprint and the 4-m sled push were measured with photoelectric cells (Brower Timing Systems, Draper, UT) for each set. The players were instructed not to consume caffeine during the 48 hours before the test. Subjects were familiarized with the test at the beginning of the season (September) before starting the experiment.

To avoid the pacing strategy, each participant performed 2 trials of 20-m sprints with maximal intensity at the end of a warm-up, which was composed of 8 minutes of muscular work and moderate-intensity running followed by 7 minutes of incremental accelerations. The minimal time was used as the sprint reference. To ensure that the player produced a maximum effort from the first sequence of the RHIEa test, a tolerance of 5% of the time reference was accepted. Otherwise, the RHIEa test was stopped and restarted after 5 minutes of passive recovery. During the test, the participants were instructed to perform each effort with maximal intensity under strong vocal encouragement and received oral feedback on their performance in each sprint and sled push.

Six indicators were computed from the RHIE test performance. The average velocity over the 10 sprints of 20 m (ASV: average sprint velocity in $\text{m}\cdot\text{s}^{-1}$) was calculated as follows: $\text{ASV} = 20 / (\text{mean of 10 sprint times})$. The average velocity over the 10 sled pushes of 4 m (ASPV: average sled push velocity in $\text{m}\cdot\text{s}^{-1}$) was calculated as follows: $\text{ASPV} = 4 / (\text{mean of 10 sled push times})$. The sprint velocity decrement (Sdec in %) was computed as follows:

$$\text{Sdec} = \left(\frac{(\text{S1} + \text{S2} + \text{S3} \dots + \text{S10})}{10 \times \text{Sbest}} - 1 \right) \times 100$$

(with S1–S10 correspond to sprint 1 to sprint 10 times and Sbest corresponds to the best sprint time). The sled push velocity decrement (SPdec) in percentage was calculated as follows:

$$\text{SPdec} = \left(\frac{(\text{P1} + \text{P2} + \text{P3} \dots + \text{P10})}{10 \times \text{Pbest}} - 1 \right) \times 100$$

(with P1–P10 correspond to sled push 1 to sled push 10 times and Pbest corresponds to the best sled push time). Two composite RHIEa indicators were proposed from these data, namely, the RHIEa score, which was defined as $\text{ASV} \times \text{ASPV}$, and the RHIEa decrement score (RHIEa dec), which was defined as

$$\text{RHIEa dec} = \frac{(\text{Sdec} + \text{SPdec})}{2}.$$

In-Game Match Data. Only data from players who participated on the field for at least 30 minutes in each game were included in the study. Velocity and acceleration data were measured using GPS units (an integrating accelerometer and a gyroscope) during the 6 games of the pretraining period and the 7 matches of the posttraining period. The time and position were recorded using a GPS unit (sampling rate: 10 Hz) fixed between the scapulae. Accelerations were recorded with a GPS-integrated triaxial accelerometer (sampling rate: 1 KHz). The total distance per minute (Dtot, $\text{m}\cdot\text{min}^{-1}$), number of runs over 80% of the participant's maximum velocity per minute (Stot in $\text{nb}\cdot\text{min}^{-1}$), and number of accelerations higher than $2.5 \text{ m}\cdot\text{s}^{-2}$ per minute (Atot in $\text{nb}\cdot\text{min}^{-1}$) were obtained from the software provided by the manufacturer (Openfield 3.4.0, Catapult, Melbourne, Australia).

During each of these official games, individual player activities were evaluated by video analysis (Sportscode, Jeddah, Saudi Arabia). Regardless of success or failure, all participations in tackle attempts, offensive and defensive rucks, mauls, scrums, and carry events attempted with the ball in hand leading to a contact were counted for each player. The numbers of occurrences were summed and expressed relative to the player's playing time on the field to define the contact rate. These data were recorded by a single observer who was unaware that his coding would be used in the study. The intrasession reliability was checked using the intraclass correlation coefficient (ICC; confidence interval at the 95% confidence level) for all events for the contact rate. The ICC values were 0.998 (0.995; 0.999) for tackles, 0.998 (0.996; 0.999) for rucks, and 1 for scrum, maul, and duel events. Then, all data

from games 1–6 in the pretraining period and from games 15–21 in the posttraining period were averaged over the corresponding period and used for the subsequent analysis.

Statistical Analyses

All data are expressed as mean $\pm$ SD. The Shapiro-Wilk test was performed to verify the normality of the variables. One-way analyses of variance (ANOVAs) were applied to determine the effect of *group* (CONT vs. RSE vs. RHIE) on external load and playing time. Furthermore, 2-way ANOVAs with repeated measures were performed to determine the effects of training (*group*: CONT vs. RSE vs. RHIE), *time* (pretraining vs. posttraining), and *group* $\times$ *time* interaction on running and rugby performance. If the ANOVAs revealed significant effects, Tukey's post hoc test was applied, Cohen's *d* effect sizes (absolute values: 0.2, 0.5, and 0.8 for small, medium, and large effects, respectively) were calculated to determine the magnitude, and Smallest Worthwhile Change (SWC), coefficient of variation (CV), and 2 coefficients of variation (2CV) were computed to assess practical relevance of changes, with trivial, possible, and certain changes, respectively (see Table 3, Supplemental Digital Content 1, <http://links.lww.com/JSCR/A459>). All statistical tests were performed using the Statistica 13.2 software package (StatSoft Inc., Tulsa, OK), and the level of significance was set at $p \leq 0.05$.

Results

Anthropometrics, Training Load, and Playing Time

No significant differences in age ($p = 0.15$), height ($p = 0.69$), body mass ($p = 0.81$), or four-skinfold body fat ($p = 0.23$) were observed between the 3 groups (Table 2). Total distance over the training period ($p = 0.26$), total PlayerLoad ($p = 0.36$), and playing time ($p = 0.24$) were similar for the 3 groups.

Repeated High-Intensity Effort Ability Test

Regarding the RHIEa test results (Figure 3, see Table 1 Supplemental Digital Content 1, <http://links.lww.com/JSCR/A459>), ANOVA revealed significant effects of *group* $\times$ *time* interaction on ASV ($F_{(2,36)} = 3.5$, $p = 0.036$), ASPV ($F_{(2,36)} = 6.81$, $p = 0.003$), and RHIEa ($F_{(2,36)} = 8.92$, $p < 0.001$) score. Post hoc comparisons only revealed that ASV, ASPV, and RHIEa score significantly increased between the pretraining and posttraining periods in the RHIE ($d = -0.39$, -0.81 , and -0.72 , respectively, $p < 0.05$) and RSE ($d = -0.53$, -0.48 and -0.60 , respectively, $p < 0.05$) groups but not in the CONT group. In posttraining period, no significant differences were observed for any variables between RHIE and RSE.

Although changes in ASV were certain in both RSE and RHIE groups, the effect size value was larger in the RSE group ($d = -0.53$, medium effect) than in the RHIE group ($d = -0.39$, small effect). Conversely, although changes in ASPV and RHIEa were, respectively, possible and certain in both RSE and RHIE groups, the effect size value of training-induced increases of ASPV and RHIEa score were greater in the RHIE group than in the RSE group ($d = -0.81$, large effect and $d = -0.48$, medium effect for ASPV, respectively; and $d = -0.72$, medium effect and $d = -0.60$, medium effect for RHIEa score, respectively).

No significant effect of *group* was observed for Sdec ($F_{(2,36)} = 1.89$, $p = 0.17$), Pdec ($F_{(2,36)} = 0.96$, $p = 0.39$), and RHIEa dec ($F_{(2,36)} = 1.74$, $p = 0.19$). A significant effect of *time* was observed for Sdec ($F_{(2,36)} = 7.04$, $p = 0.012$) and RHIEa dec ($F_{(2,36)} = 5.54$, $p = 0.024$) but not for Pdec ($F_{(2,36)} = 2.44$, $p = 0.13$). Posttraining Sdec and RHIEa dec were significantly lower than the respective pretraining values ($p < 0.05$). In addition, the effect of RHIE training on RHIEa dec was large ($d = 0.825$) and tended to be significant ($p = 0.063$).

High-Intensity Efforts in Game

No significant *group* effect was observed concerning Stot ($F_{(2,36)} = 0.93$, $p = 0.41$), Dtot ($F_{(2,36)} = 0.58$, $p = 0.56$), Atot ($F_{(2,36)} = 1.52$, $p = 0.23$), and contact rate ($F_{(2,36)} = 0.39$, $p = 0.68$).

Regarding all of the in-game high-intensity effort variables (Figure 4, see Table 2 Supplemental Digital Content 1, <http://links.lww.com/JSCR/A459>), ANOVA revealed significant *group* $\times$ *time* effects on Stot ($F_{(2,36)} = 8.127$, $p = 0.002$), Dtot ($F_{(2,36)} = 8.93$, $p < 0.001$), Atot ($F_{(2,36)} = 23.71$, $p < 0.001$), and contact rate ($F_{(2,36)} = 41.69$, $p < 0.001$). Post hoc comparisons revealed that all high-intensity effort in game variables significantly increased between the pretraining and posttraining periods in the RHIE ($d = -0.28$, -0.50 , -0.37 and -0.97 , respectively, $p < 0.05$) and in the RSE ($d = -0.47$, -0.38 , -0.32 and -0.28 , respectively, $p < 0.05$) groups, whereas Atot significantly decreased ($d = 0.91$, $p < 0.001$, large change) between the pretraining and posttraining periods in the CONT group. In posttraining period, no significant differences were observed for any variables between RHIE and RSE.

Although changes in Dtot were certain in RHIE group and RSE group, the effect size value was greater in the RHIE group than in the RSE group ($d = -0.50$, medium effect and $d = -0.38$, small effect, respectively). Furthermore, although changes in contact rate, Atot, and Dtot were trivial for both RSE and RHIE groups, the effect size value in contact rate was more than twice in RHIE training than in RSE training ($d = -0.97$, large effect and $d = -0.28$, small effect, respectively), whereas effect size value of Stot was lower in RHIE training compared with RSE training ($d = -0.28$, small effect and

Table 2

Mean ($\pm$ SD) of anthropometric, training load, and playing time characteristics for the repeated high-intensity exercise (RHIE), repeated sprint exercise (RSE), and control (CONT) groups.

<i>n</i>	RHIE 16	RSE 15	CONT 8	<i>F</i>	<i>p</i>
Age (y)	28.4 $\pm$ 3.4	25.9 $\pm$ 4.1	25.6 $\pm$ 4.6	$F_{(2, 36)} = 1.96$	0.15
Height (cm)	183.3 $\pm$ 9.4	185.1 $\pm$ 8.1	182.4 $\pm$ 5.3	$F_{(2, 36)} = 0.36$	0.69
Body mass (kg)	97.5 $\pm$ 16.8	97.5 $\pm$ 16.8	99.7 $\pm$ 11.9	$F_{(2, 36)} = 0.21$	0.81
Four-skinfolds body fat (%)	15.9 $\pm$ 4.4	15.2 $\pm$ 3.5	17.4 $\pm$ 4.3	$F_{(2, 36)} = 1.50$	0.23
Average playing time (min)	643 $\pm$ 185	699 $\pm$ 130	589 $\pm$ 92	$F_{(2, 37)} = 1.46$	0.24
Total distance on training period (m)	82,368 $\pm$ 34,215	76,561 $\pm$ 34,924	59,055 $\pm$ 21,495	$F_{(2, 37)} = 1.39$	0.26
Total PlayerLoad on training period (A.U.)	7,957 $\pm$ 3,149	7,293 $\pm$ 3,754	5,932 $\pm$ 2,210	$F_{(2, 37)} = 1.03$	0.36

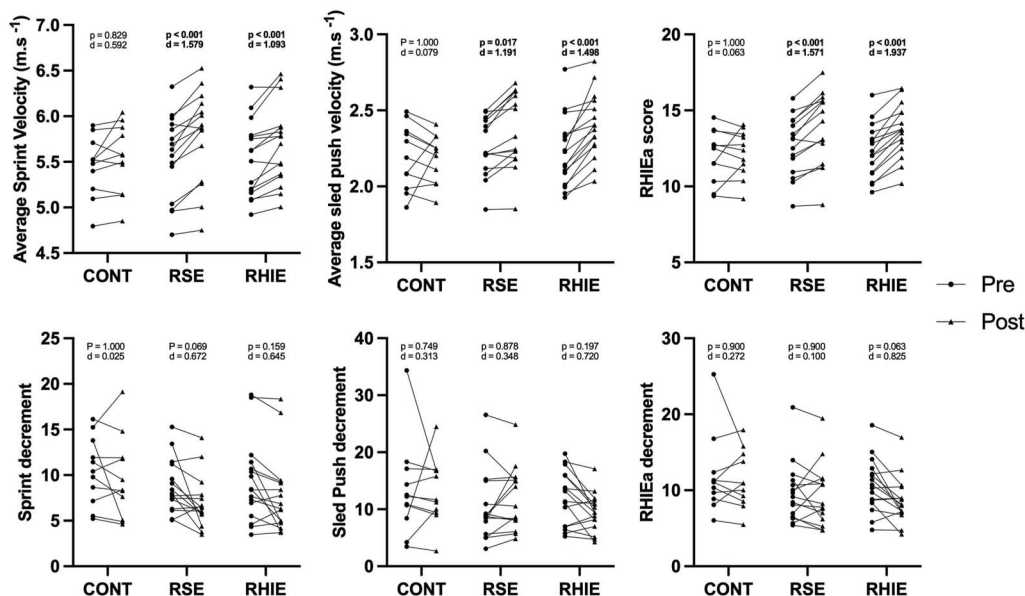


Figure 3. Performance indices in the repeated high-intensity effort (RHIE) ability test between pretraining and posttraining periods in the control (CONT), repeated sprint exercise (RSE), and RHIE training groups.

$d = -0.47$, small effect, respectively). The effect size of value in Atot was similar in RHIE training and RSE training ($d = -0.37$ and $d = -0.32$, small effects respectively).

Discussion

The aim of this study was to investigate the effects of RHIE training and RSE training on RHIEa and in-game high-intensity

efforts in professional rugby union players. The main finding was that both the RHIE and RSE interventions were efficient to improve RHIE ability and high-intensity efforts in game. Our results also show that effect size values concerning the effects of training on Stot and ASV are higher in RSE than in RHIE, suggesting that gains could be higher for RSE. Conversely, gains in resisted and contact efforts (contact rate, ASPV, RHIEa score) could be higher for RHIE than for RSE. A previous study demonstrated that gains

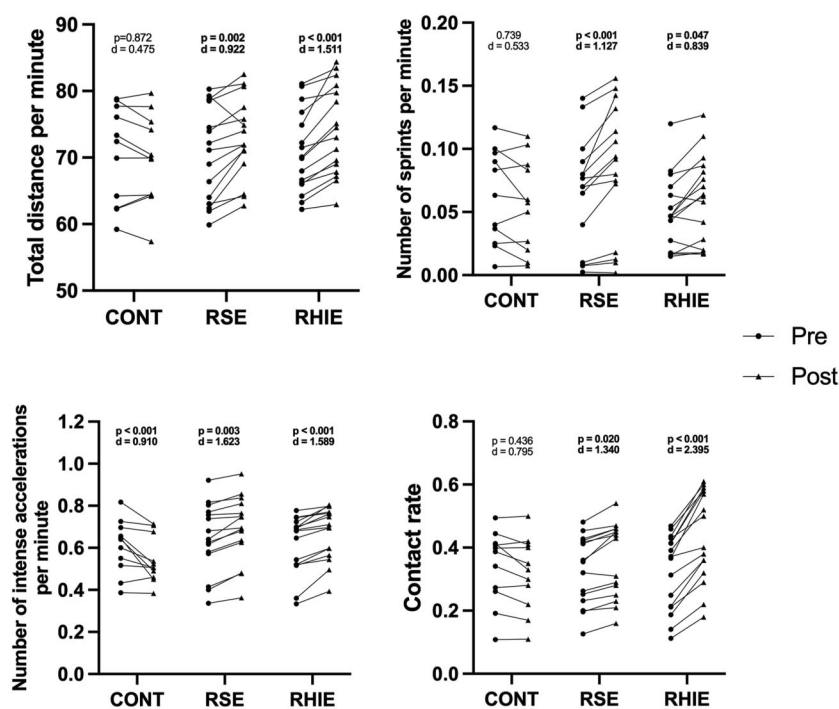


Figure 4. In-game high-intensity effort between pretraining and posttraining periods in the control (CONT), repeated sprint exercise (RSE), and repeated high-intensity effort (RHIE) training groups. RHIE = repeated high-intensity effort.

in RSa were obtained after sprint repetition training (3). Thus, as demonstrated by Buchheit et al. (7), RSa significantly improves after a repeated sprint training program by both reducing the average sprint time and the best sprint time. Similar results were observed in our study, with a 4.14% mean improvement in ASV after RSE training. However, whether RSE training could improve the RHIEa in rugby union was unknown. Noteworthy, RSE training resulted in increased ASPV and RHIE scores (+4.41% and +9.08%, respectively), which highlights that such training was also effective at improving several determinants of RHIEa. Although the effects of RSE training methods have been extensively studied in team sports (3,6), no longitudinal study had demonstrated that RSE training could improve players' performance in in-game high-intensity efforts in rugby union. Our results indicated that RSE training induced significant increases in in-game high-intensity efforts, including higher distance and contact rate and more sprints and accelerations. These results are consistent with those of Glaise et al. (23), who demonstrated that increases in RSa levels correlated with the number of sprints, accelerations, and contact situations in rugby union. Repeated sprint ability has also been related to a higher activity ratio (i.e., the number of actions per time unit) (37) and to the number of RHIE bouts produced in rugby league games (21), confirming that the development of this physical ability may be a determinant of rugby union performance.

Literature on the effects of RHIE training on RHIEa and in-game performance is highly limited. Our results revealed that ASV, ASPV, and RHIE score were significantly higher after RHIE training (+3.07%, +8.11%, and +11.13%, respectively). These results agree with those of a previous study (39), which demonstrated that RHIEa could be improved using a HIIT rugby block preceded by a preconditioning contact sequence or a preconditioning running sequence, with significant improvements in total sprint time and 20-m sprint time. Our results also indicated that RHIE training significantly increased in-game high-intensity efforts. These results are consistent with those of Vachon et al. (39), who revealed that the number of sprints at $\geq 90\%$ of the maximum velocity of a player in a rugby union game was significantly increased after 4 weeks of a HIIT training block, including preconditioning running and contact sequences. Our results thus reinforce the statement that RHIEa may directly influence the performance in in-game high-intensity efforts, which can immediately affect the success of an action (10) or a game (12,21).

Whether RSE and RHIE training differentially impact RHIEa and in-game high-intensity efforts was unknown. Although our results indicated that these 2 training methods have positive effects on RHIEa and in-game efforts, differences were observed between the 2 training methods when considering the relative improvements in RHIEa. Although effect size value in ASV was higher in the RSE group than in the RHIE group, suggesting a larger improvement in RSE group, effect size values in ASPV was larger in the RHIE group compared with the RSE group, suggesting a larger improvement in RHIE group. These results could be explained by the different nature of the efforts used in the training programs. Repeated sprint ability and RHIEa should be considered 2 distinct physical abilities because they can differently influence performance (26). As previously demonstrated (27,28), contact efforts (which are included in RHIE training) generate additional fatigue, which strongly affects players' running intensities; conversely, significant improvements in RSa but not in intense neuromuscular efforts were found after repeated sprint shuttle training (6). The effects of including preconditioning combat or preconditioning running before high-intensity interval

training on RHIEa were recently compared (40): Sdec improvements were higher in the study's preconditioning sprint group compared with its preconditioning combat group. Our results demonstrated a large effect, although a nonsignificant one, of RHIE training on RHIEdec ($d = 0.83$, $p = 0.063$). This absence of a significant difference could be explained by high individual variability. Moreover, the absence of relationships with performance decrements could be related to the indicator's poor reliability (33). In addition, our results partly confirmed those of Fornasier-Santos et al. (16), who did not measure any significant improvement in the number of sprints (i.e., $>85\%$ max velocity) performed after repeated sprint training in normoxia; conversely, a significantly higher number of sprints (i.e., $>85\%$ max velocity) was observed after repeated sprint training in hypoxia. Taken together, our results suggest that RHIEa can be improved in different components depending on the chosen training method. In rugby union, as large differences exist in activity profiles according to the player's playing position (34) and profile, we suggest that practitioners should choose a method of training that is relevant to the requirements of each player.

Furthermore, the magnitude of the effects of the RSE and RHIE methods also appear as potentially different concerning high-intensity efforts in rugby union games. Although the effects size values of RSE and RHIE training on accelerations were similar, RHIE training led to a larger magnitude of effect in total distance compared with RSE training. More interestingly, although effects size values in the number of sprints was greater after RSE training than after RHIE training, increases in training-induced contact rate were almost triple after RHIE training compared with RSE training. These results suggest that RHIE training could be more effective at increasing the occurrence of combat actions, whereas RSE could be more effective at increasing the occurrence of sprints. In line with previous publications, RHIE training can be considered to possibly (a) improve the ability of players to repeat contact efforts and thus (b) directly influence the success of a match (35), as winning teams in Super Rugby perform more tackles than losing teams. Furthermore, because the contact phases concern all players (11) and particularly forwards (9), it could be relevant to propose RHIE training to the players according to their position and requirements. If the aim is to improve the RSa (which could be more relevant for backs), RSE training may be more beneficial than RHIE. Conversely, for players who require more combat sequences (which is the case for forwards), RHIE training could be more suitable because the ability to repeat sled push-type efforts (inducing higher levels of strength for a lower-velocity condition) is improved more by RHIE training.

The different nature of efforts performed in RSE and RHIE programs induce specific physiological adaptations (32). Repeated high-intensity effort training uses different types of efforts, particularly contact and running efforts, which lead to mixed fatigue related to both the perturbation of muscle activation and metabolic fatigue (32). Such training allows players to limit performance decrements in intense neuromuscular efforts (e.g., the sled push) and thus to be less affected by the repetition of contact efforts in game (as observed with the significant reduction of SPdec for RHIE training in our study). By contrast, RSE training, exclusively based on high-intensity running repetition, leads to metabolic fatigue related to the decrease in PCr, inorganic phosphate accumulation, and H^+ increase (30). This method allows players to increase their average velocity in high-intensity efforts, such as sprints, which facilitates improvements in acceleration and sprinting repetition in rugby union games. One can

hypothesize that physiological adaptations to RHIE and RSE training could also lead to higher levels of horizontal strength in sprinting and contact actions, which would make players more efficient in contact bouts (24). One can also hypothesize that although RSE training allows players to improve their RHIEa by improving their capacities to produce and repeat intense running efforts, RHIE training improves the ability to repeat contact efforts, which could make players' running efforts less limited by contact-induced fatigue.

One of the main limitations of this study was that it did not evaluate the technical dimensions provided by the different training programs. Indeed, the contact efforts that were part of the RHIE training program could have improved the players' technical abilities, which could have influenced their ability to use their energy during contact bouts and thus to achieve more in game. Another important limitation concerns the use of the sled to simulate a contact force. Although contact forces generate a horizontal thrust constraint and do not systematically go through the ground, the sled thrust does not include contact with the upper body or passage through the ground, which partially simulates the totality of a contact force. For future studies, any further questioning is required to measure contact efforts performance as accurately and reliably as possible. Furthermore, as the relevance of the changes is similar between the RSE and RHIE groups, the results obtained from the magnitude of changes analysis are interpreted as trends. It is necessary to repeat these analyses in the future to acquire more solid statements.

Moreover, although a large sample of games was used, it cannot be completely excluded that the results could have been influenced by the level of the opponents (19) and the competition level (15). Thus, the results should not be generalized to other levels of rugby practice or other types of rugby. In addition, as our results were obtained from one team, it is possible that the results might have been different with teams practicing a different style of play.

Practical Applications

This study has demonstrated that the RSE and RHIE methods are effective for developing RHIEa and players' activity in rugby union games. Particularly, RSE training would be more effective to improve the number of sprints and accelerations produced in a game, whereas RHIE training would be more beneficial to increase the number of contacts and the intensity of movement. As the gains in some abilities and game performance depend on the training method administered, we suggest that strength and conditioning coaches choose the most appropriate method according to the player's profile and playing position as well as the team's playing strategies.

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